

Effects of a 10-d Military Field Exercise on Body Composition, Physical Performance, and Muscle Cells in Men and Women

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ABSTRACT

VIKMOEN, O., H. K. TEIEN, R. TANSØ, A. AANDSTAD, E. LANDER, K. T. CUMMING, S. ELLEFSEN, K. HELKALA, and T. RAASTAD. Effects of a 10-d Military Field Exercise on Body Composition, Physical Performance, and Muscle Cells in Men and Women. *Med. Sci. Sports Exerc.*, Vol. 56, No. 4, pp. 682–696, 2024. **Purpose:** This study aimed to investigate the effects of a demanding military field exercise on physical performance, body composition, and muscle cellular outcomes in men and women. **Methods:** Ten men (20.5 ± 0.5 yr) and 8 women (21.4 ± 1.4 yr) completed a 10-d field exercise consisting of extensive physical activity with food and sleep restriction. Acquisition of body composition, physical performance, blood, and muscle biopsies samples were done before and 1, 7, and 14 d after the exercise. **Results:** There were no sex differences in the response to the exercise. Body mass was decreased with $5.6\% \pm 1.8\%$ and fat mass with $31\% \pm 11\%$ during the exercise. Both were still reduced after 14 d ($2.5\% \pm 2.3\%$, $P < 0.001$, and $12.5\% \pm 7.7\%$, $P < 0.001$, respectively). Isometric leg strength did not change. Peak leg extension torque at 240°s^{-1} and counter movement jump height were reduced with $4.6\% \pm 4.8\%$ ($P = 0.012$) and $6.7\% \pm 6.2\%$ ($P < 0.001$), respectively, and was still reduced after 14 d ($4.3\% \pm 4.2\%$, $P = 0.002$, and $4.1\% \pm 4.7\%$, $P = 0.030$). No changes occurred in fiber CSA, fiber types, proteins involved in calcium handling, or HSP70. During the exercise, α B-crystallin levels decreased by $14\% \pm 19\%$ ($P = 0.024$) in the cytosolic fraction and staining intensity on muscle sections tended to increase ($17\% \pm 25\%$, $P = 0.076$). MuRF1 levels in the cytosolic fraction tended to decrease ($19\% \pm 35\%$) and increased with $85\% \pm 105\%$ ($P = 0.003$) in the cytoskeletal fraction 1 wk after the exercise. **Conclusions:** The field exercise resulted in reduced body mass and physical performance in both sexes. The ability to produce force at high contraction velocities and explosive strength was more affected than isometric strength, but this was not related to any changes in fiber type composition, fiber area, Ca^{2+} handling, or fiber type-specific muscle damage. **Key Words:** MUSCLE STRENGTH, RECOVERY, MUSCLE DAMAGE, MUSCLE FIBERS, HEAT SHOCK PROTEINS, CALCIUM

Operational training and demanding military field exercises are a common and an important part of military education and training. Such exercises often involve very high activity levels and energy expenditure combined with low energy intake and lack of sleep (1,2). These factors lead to energy deficits that have been reported to be as high

as $5000\text{--}6000 \text{ kcal}\cdot\text{d}^{-1}$ over a 3- to 7-d period (1,3–5), further leading to large losses of body mass, body fat, and muscle mass (2,4,6,7). These exercises have also been shown to negatively affect different aspects of physical performance, including aerobic power (3), anaerobic performance (8), maximal and explosive muscle strength (2,6,7,9), and performance in military specific physical tasks (6). In addition, the recovery of physical performance from such exercises seems to be slow as we have recently reported that explosive strength was not recovered within 2 wk (4,10), and not within 6 wk after more prolonged exercises (11).

Reductions in physical performance may have negative consequences for mission success, and therefore it is important to come up with countermeasures to prevent the reduction in physical performance and/or to aid recovery. However, in a recent review, O'Hara et al. (12) concluded that documentation for effective countermeasures against the negative effect of demanding military training on performance is lacking in the

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literature. Some have tried to increase the protein intake during field exercises, without any beneficial effect (13,14). Consequently, better knowledge on the physiological mechanisms behind the performance impairments will be important to design effective countermeasures against the negative effects of operational training and field exercises. These mechanisms remain unclear, and there are multiple factors that can contribute to the observed reductions in performance. Previous studies have speculated that the changes in body composition and endocrine changes can explain the decline in physical performance (2,3,7). However, there are only small to no correlations between the changes in lean body mass, endocrine changes, and changes in performance after demanding military field exercises (4,7,11,15). In a previous study from our group, all changes in body composition and hormones were recovered within 1 wk, although physical performance was reduced for more than 2 wk. Furthermore, despite men losing a substantial amount of muscle mass while women maintained their muscle mass during the exercise, there were no sex differences in performance changes. This demonstrates a discrepancy between changes in body composition, including muscle mass, and physical performance (10). Therefore, the mechanisms behind the reduced performance is probably more related to changes in muscle quality rather than muscle mass. Muscle damage related to the high activity levels is a possible mechanism, and reports of large increases in serum creatine kinase (CK) in both men and women after such exercises (4,10) certainly indicate that some muscle damage exists. However, a clear relationship between individual changes in CK levels and changes in performance has not been established (10).

We have previously reported that explosive strength and maximal power development are more affected and recover slower than maximal isometric strength in both men and women (4,10). Therefore, it can be speculated that the underlying mechanisms behind the decline in muscle performance involve structural damages or disturbances to energy systems that are important for explosive strength rather than maximal strength. Because type II fibers have higher maximal contraction velocity than type I fibers, one possible mechanism could be that military field exercises negatively affect type II fibers to a larger degree than type I fibers with for example larger muscle damage in type II fibers or with a fiber type shift toward slower fiber types. A typical response to muscle damage is increased expression and accumulation of heat shock proteins (HSP) during the recovery period. These proteins have previously been reported to bind and accumulate to areas within the cell with mechanical damage in both sexes (16,17). The levels of HSP accumulated in muscle fibers correlate with the decline in force generating capacity after exercise (18,19). Thus, HSP can give an indication to what degree muscle damage exists and which muscle fiber type is affected by a demanding military field exercise.

Another possible mechanism might be related to impaired calcium (Ca^{2+}) handling and dynamics in the muscle cells. For example, prolonged periods of high activity and endurance training have been reported to lead to a dissociation of

the channel stabilizing protein calstabin1 from the Ca^{2+} release channel, the ryanodine receptor (RyR1) (20,21). This dissociation can result in leaky channels at rest, impaired Ca^{2+} release, and therefore impaired exercise capacity, muscle function, and fatigue resistance (20). Furthermore, in male participants, six 30-s all-out cycling bouts have been reported to lead to an extensive fragmentation of RyR1 also causing them to become leaky at rest leading to increased resting cytosolic Ca^{2+} , impaired Ca^{2+} release, and depressed muscle force, especially at lower stimulation frequencies (21). Unfortunately, no previous study has investigated fiber type-specific changes or changes in Ca^{2+} handling after demanding military field exercises.

The majority of previous studies investigating the effects of demanding short-term military field exercises have only included male participants. Previous studies have reported that men have a higher absolute energy expenditure and energy deficit compared with women during field exercises. However, when adjusted for body mass or fat-free mass, no sex differences exists (5,10,22). It has also been reported that women oxidize more fat (5) and loses less muscle mass (5,10) during such short-term field exercises compared with men. We have also observed that the recovery of vertical jump performance after a short but extremely demanding military field exercise differed between men and women despite similar physical performance decrements during the exercise (10). Therefore, it might be speculated that the physiological mechanisms behind changes in performance during and in the recovery phase from such short-term exercises differ between the sexes. However, these have not been investigated.

Therefore, the main aim of the current study was to investigate how different aspects of physical performance are affected by and recovers from a demanding short-term military field exercise. A special focus in this study was to investigate the underlying physiological and cellular mechanisms behind the decrements in physical performance and investigate any sex differences in these changes.

The main hypotheses were that factors related to explosive strength and maximal power would be more affected by a demanding field exercise than maximal strength. These changes would be related to more pronounced impairments in type II fibers compared with type I fibers, and that changes in calcium-handling proteins are involved in this etiology. The impairments in explosive strength and power together with the muscle cellular outcomes were expected to recover faster in women than men.

METHODS

Participants

Ten men (20.5 ± 0.5 yr, 181 ± 6.7 cm, and 76.7 ± 6.4 kg) and 8 women (21.4 ± 1.4 yr, 166 ± 3.1 cm and 65.4 ± 12.3 kg) were recruited from second year military students at the Norwegian Defense Cyber Academy. In May each year, these students participate in a 10-d long mandatory demanding military field exercise called “cyber endurance.” Participation in the study was voluntary, and all participants were free to withdraw at

any time. All students performed the mandatory field exercise regardless of their participation in the research study, and participation in the study did not affect the content of the field exercise. The project was evaluated by the Norwegian Regional Committees for Medical and Health Research Ethics. All participants provided a written informed consent, and the study was performed in compliance with the Declaration of Helsinki.

Experimental Design

Blood sampling, measurements of body composition, and testing of physical performance were performed before (pre) and 24 h (post 24 h), 1 wk (post 1 wk), and 2 wk (post 2 wk) after the cyber endurance exercise (Fig. 1). In addition, muscle biopsies from vastus lateralis were collected approximately 3 wk before the exercise (pre), at post 24 h and at post 1 wk. To reduce the numbers of biopsies the participants had to undergo, no biopsy was taken at post 2 wk. On each test day, blood samples were collected, and body composition measurements were performed in the morning before breakfast (between 0600 and 0800 h). The physical tests were performed 2–5 h after breakfast. Biopsies were collected between breakfast and physical tests. All participants performed their individual testing at the same time of day (± 1 h) each test day.

Procedures and Protocols

Cyber endurance exercise. The cyber endurance exercise lasted for 10 d and consisted of a high degree of various types of physical activity, sleep deprivation, and low energy intake to create a realistic and demanding operational environment. The exercise involved soldiers performing physically and mentally challenging military tasks in an outdoor environment. The exercise encompassed a range of activities, including cyber-specific tasks, marching, patrolling, and physical combat conditioning training, which lasted for several hours. Additionally, the soldiers had to carry their personal combat equipment weighing approximately 20 kg throughout the entire exercise. The exercise was designed to gradually decrease the sleep, rest, and food and gradually increase the physical

and mental demands over the course of 10 d. During the three first days, the soldiers ate as normal and slept in the military camp. During this phase, they had physical training in the morning between 0600 and 0800 h before they ate breakfast. This physical training consisted of intense combat simulation training and included exercises such as carrying ammo boxes, carrying fellow soldiers, lifting and turning big tires, agility training, push-ups, burpees, etc. The rest of the day, they had a mix of classroom lessons and more physical demanding activities such as marching and patrolling. In these 3 d, they slept between 0000 and 0530 h. The rest of the exercise, they slept out in the field and operational activity lasted for a minimum of 18 h each day. The soldiers slept approximately 2 to 3 h each night. During this phase, the soldiers performed cyber-specific engineering tasks together with physical tasks, such as marching, patrolling, and combat conditional training (as described above) lasting for several hours each day. Both men and women received field rations containing approximately 900 kcal·d⁻¹, and they had no access to outside food. After the exercise, both the officers and the soldiers confirmed that all field rations were eaten. During the exercise, the students were divided into mixed-sex teams so both men and women performed the same exercises. However, each team solved the challenges as they wanted, and we have no control on how each team distributed different tasks, for example, distribution of the load carried. Accelerometers (ActiGraph wGT3X-BT; ActiGraph, Pensacola, FL), worn on the nondominant wrist, were used to estimate the activity energy expenditure with the Freedson VM3 Combination algorithm in the Actilife software (ActiGraph wGT3X-BT, ActiGraph). Basal metabolic rate was calculated with the formula $BMR = 21.6 \times LBM \text{ (kg)} + 370$ (23). Total energy expenditure was the sum of activity energy expenditure and basal metabolic rate. In absolute values, the average energy expenditure per day during the exercise was significantly larger in men ($5775 \pm 558 \text{ kcal} \cdot \text{d}^{-1}$) compared with women ($4717 \pm 644 \text{ kcal} \cdot \text{d}^{-1}$, $P = 0.002$). However, normalized for body mass, there were no sex differences (men, $75.5 \pm 5.7 \text{ kcal} \cdot \text{kg}^{-1} \cdot \text{d}^{-1}$; women, $72.9 \pm 6.8 \text{ kcal} \cdot \text{kg}^{-1} \cdot \text{d}^{-1}$; $P = 0.396$). Both sexes had similar energy expenditure from

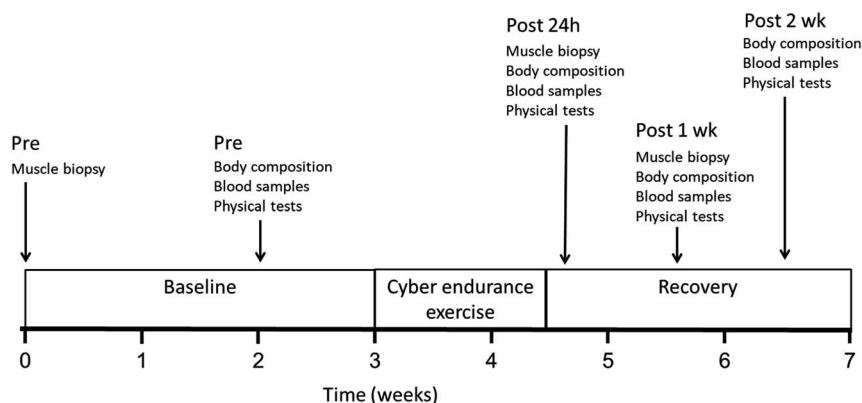


FIGURE 1—Timeline over the study. The arrows represent when tests were performed. Three weeks before the cyber endurance exercise a resting muscle biopsy from vastus lateralis was collected. One week before the exercise, measurements of body composition and physical performance were done, and venous blood was collected. All these tests were repeated 24 h (post 24 h), 1 wk (post 1 wk), and 2 wk (post 2 wk) after the exercise, except that muscle biopsy was not collected at post 2 wk.

day to day, ranging from 5350 to 5990 kcal·d⁻¹ for men and from 4640 to 5011 kcal·d⁻¹ for women. With 900 kcal·d⁻¹ of energy intake, the estimated energy deficit during the last 6 d was larger in men (4899 ± 574 kcal·d⁻¹) compared with women (3819 ± 684 kcal·d⁻¹, *P* = 0.002). Normalized for body mass, this was 64.0 ± 5.8 kcal·kg⁻¹·d⁻¹ for men and 58.6 ± 5.3 kcal·kg⁻¹·d⁻¹ for women (*P* = 0.063). The scientific team had no influence on the content of the exercise that was controlled solely by the Norwegian Defense Cyber Academy.

Body composition. Body composition and body mass were measured with bioelectrical impedance analysis on an InBody 720 machine (Biospace Co., Seoul, Korea) according to the manufacturer's instructions. The bioelectrical impedance analysis was administered after an overnight fast between 0600 and 0800 h. The students were instructed not to shower or be physically active from the time they woke up until the test was completed and told to go to the toilet before the measurements. The participants also stood upright the last 5 min before the test. Participants performed all measurements in their underwear. Compared with DXA measurements, the InBody 720 has been shown to systematically underestimate fat mass by 3.6 ± 1.3 kg (4) and has been reported to have an average test–retest measurement error in percent body fat of 2.3 and 2.6 percentage points and an ICC of 0.97 and 0.98 in men and women, respectively (24).

Blood sampling. Blood samples were collected in the morning (0600–0800 h) in a fasted state to investigate anabolic and catabolic hormones and markers of muscle damage. The students were instructed not to be physically active in the period between they woke up, and the blood was collected. Whole blood was mixed before clotting in room temperature for 30 min and centrifuged at 1300g for 10 min. Serum was transferred into tubes and stored frozen at –80°C until analyses. Serum was analyzed for cortisol (analytic CV, 6.8%), testosterone (7.0%), CK (3.9%), at Furst Medical Laboratory (Oslo, Norway), and IGF1 (6.0%) at the Hormone Laboratory (Oslo, Norway).

Muscle Biopsy Analyses

Muscle biopsies were sampled from the midportion of the vastus lateralis under local anesthesia (Xylocainadrenalin, 10 mg·mL⁻¹ + 51 g·mL⁻¹; AstraZeneca PLC, London, UK), with a modified Bergstrom technique. Samples intended for immunohistochemistry were aligned and covered in O.C.T embedding matrix (CellPath, Newtown, UK) and frozen in isopentane cooled in liquid nitrogen. Samples for Western blotting were gently washed in cold physiological saline and fat, connective tissue, and blood were discarded before they were weighed and snap frozen in liquid nitrogen. All samples were transferred to storage at –80°C for subsequent analysis.

Immunohistochemistry. Eight-micrometer-thick cross sections were cut at –20°C using a microtome (CM3050; Leica Microsystems GmbH, Wetzlar, Germany) and mounted on microscope slides (Superfrost Plus; Thermo Fisher Scientific,

Inc., Waltham, MA). The sections were then air-dried and stored at –80°C until further analyses.

After being thawed, the muscle sections were blocked for 60 min with 1% BSA (A4503; Sigma-Aldrich Corp., St. Louis, MO) in a 0.05% PBS-T solution (524650; Calbiochem, EMD Biosciences, Inc., San Diego, CA).

One slide of the muscle cross sections was incubated with antibodies against MyHCI (1:500; DSHB Hybridoma Product BA-D5, deposited by Schiaffino, S) and dystrophin (1:500; ab15277; Abcam Plc), one was incubated with antibodies against HSP70 (Cat#ADI-SPA-810, Enzo Life Sciences) and dystrophin, and one with antibodies against alpha-B crystallin (Cat#ADI-SPA-222, Enzo Life Sciences) and dystrophin at 4°C overnight followed by incubation with appropriate secondary antibodies (Alexa Fluor, A11005, or A11001, Invitrogen, Inc., Carlsbad, CA) for 1 h at room temperature. Between stages, sections were washed for 3 × 10 min in a 0.05% PBS-T solution with gentle agitation. Muscle sections were covered with a coverslip and embedded in ProLong Gold Antifade Reagent with DAPI (P36935; Invitrogen Molecular Probes, Eugene, OR) and left to cure overnight at room temperature. Muscle sections were visualized using a high-resolution camera (DP72; Olympus Corp., Tokyo, Japan) mounted on a microscope (BX61; Olympus Corp.) with a fluorescence light source (X-Cite 120PCQ; EXFO Photonic Solutions Inc., Mississauga, Ontario, Canada). To compare staining intensities between muscle sections, camera and software settings were fixed when taking micrographs of the sections stained against the HSP. HSP staining intensity was analyzed and related to each fiber type using ImageJ with Fiji image processing package (25).

Fiber type distribution and fiber cross-sectional area were identified and analyzed using TEMA software (CheckVision, Hadsund, Denmark). Fibers showing positive stain against BA-D5 was quantified as a type I fiber, whereas fibers not stained (BA-D5 negative) was quantified as a type II fiber. For representative immunohistochemistry pictures, see Supplemental Figure 1 (<http://links.lww.com/MSS/C950>).

Immunoblotting. For extraction of proteins, about 50 mg muscle tissue was mechanically homogenized by rotor-stator in 1 mL homogenizing buffer (T-PER™, Tissue Protein Extraction Reagent, 78510; Thermo Scientific, Rockford, IL) with 20 μL Halt™ protease, 10 μL phosphatase inhibitor cocktail (1861281, Thermo Scientific), and 10 μL EDTA (1861274, Thermo Scientific). Samples were subsequently mixed for 30 min at 4°C and later centrifuged at 10,000g for 10 min at 4°C. The supernatant was aliquoted to microcentrifuge tubes and stored at –80°C until further analyses.

Additionally, about 50 mg of muscle tissue was homogenized and fractionated into cytosol, membrane, nuclear, and cytoskeletal fractions, using a commercial fractionation kit according to the manufacturer's procedures (ProteoExtract Subcellular Proteo Extraction Kit, 539790, Calbiochem; EMD Biosciences GmbH, Schwalbach, Germany). All cell fractions

were aliquoted to microcentrifuge tubes and stored at -80°C until further analyses.

Total protein concentrations were measured with a commercial kit (Bio-Rad DC protein micro plate assay, 5000113, 5000114, 5000115; Bio-Rad Laboratories, Inc., Hercules, CA) and measured using a filter photometer (Expert 96; ASYS Hitech, Cambridge, UK) and calculated using the provided software (Kim Version 5.45.0.1; Daniel Kittrich).

Samples were analyzed by Western blotting. Between 4 and 25 μg of denatured proteins (depending on sample) were separated by electrophoresis by 4%–12% gradient gels (NativePAGE, BN1002BOX, Invitrogen, or NuPAGE, NP0321BOX, Invitrogen) at 200 V for 45–60 min in cold MES SDS running buffer (NuPAGE NP0002, Invitrogen). Separated proteins were then transferred to a PVDF membrane (162–0177, Bio-Rad) with XCell II Blot Module (Invitrogen) at 30 V for 90 min in cold transfer buffer (NuPAGE, NP00061) and an antioxidant (NuPAGE, NP0005). After transfer, the PVDF membranes were blocked in a 5% skim milk solution (EMD Millipore Corporation, Billerica, MA) in TBS-t (Tris-buffered saline, 170-6435, Bio-Rad, and 0.1% Tween® 20, P7949-100ML, Sigma-Aldrich) for 2 h at room temperature. After blocking, the membranes were cut into smaller portions based on the molecular weight (weight marker; Protein Ladder PS 11, 310005, GeneOn, Germany) of the appropriate proteins and incubated in primary antibodies according to Supplemental Table 1 (Overview over the antibodies used for Western blotting in the study, <http://links.lww.com/MSS/C950>) overnight at 4°C with gentle agitation on a grate board. The next day, the membranes were washed with TBS-t for 15 min and then 3×5 min in TBS before incubated in appropriate secondary antibody diluted in a 1% skim milk solution in TBS-t. After incubation with secondary antibody, the membranes were washed. Membranes were then incubated for 5 min with chemiluminescent substrate SuperSignal WestDura® (Extended Duration Substrate, 34076, Thermo Scientific). Chemiluminescence was measured using ChemiDoc™ MP Imaging System and analyzed using Image Lab™ Software (Bio-Rad).

SERCA1 and SERCA2 are exclusively expressed in muscle fiber types II and I, respectively. To account for the large sample to sample variation in muscle fiber type (26), the individual SERCA1 and SERCA2 data at each time point the values were adjusted for the fiber type composition measured in the corresponding IHC piece using the formulas:

$\text{SERCA1} = \text{SERCA1 expression measured by Western blotting} / 100 \times \text{percentage type II fibers}$

$\text{SERCA2} = \text{SERCA2 expression measured by Western blotting} / 100 \times \text{percentage type I fibers}$

Physical Performance Test

After a general warm-up, explosive strength performance of the leg muscles was tested with a counter moment jump (CMJ). This was followed by testing of maximal isometric and isokinetic torque in the leg extension dynamometer.

General warm-up. The general warm-up consisted of 10 min cycling with a low and self-selected intensity. During the last 3 min, the students performed three 30-s bouts at a higher intensity.

Maximal isometric and isokinetic torque in leg extension. Maximal isometric torque (MVC) in leg extension and peak torque during a leg extension at $240^{\circ}\cdot\text{s}^{-1}$ were tested in a dynamometer (Humac Norm; CSMi Solutions, Stoughton, MA). The participants were seated with a 90° hip angle and were stabilized in this position using chest, hip, and thigh straps. The input axis of the dynamometer was aligned with the participants' knee joint, and the ankle was strapped to a lever arm. The participants held their arms in front of their chest during all tests. First, the participants performed three maximal isometric leg extension against the lever arm with a 90° knee angle. The contractions lasted for 5 s, and a 60-s rest was given between each attempt. The participants were instructed to perform the muscle action as forcefully and quickly as possible. The attempt with the highest maximal torque was chosen for the statistical analyses. After 1 min of rest, three maximal isokinetic leg extensions from 90° knee angle to full extension were performed against the lever arm at an angular velocity of $240^{\circ}\cdot\text{s}^{-1}$. There was a 60-s rest between each attempt. The attempt with the highest torque recorded during the contraction was chosen for statistical analyses. Strong verbal encouragement was given to the participants at all attempts.

Counter moment jump. The CMJ test was performed on a force plate (HUR Labs, Tampere, Finland). The participants were instructed to stand on the platform with feet at shoulder width. After a countdown from the test administrator, the participants then completed the jump. The jump was performed with a flexion of the knee and hip joint to approximately 90° in the knee joint, followed by a rapid countermovement and extension of the lower extremities. Hands were placed on their hips throughout the entire movement. Each participant was given three attempts, with a 30-s rest between each attempt. If the third attempt was highest, additional jumps were performed. The results from highest jump were used in the statistical analyses.

Statistics

All statistical analysis was done in IBM SPSS (IBM SPSS Statistics, version 25; IBM Corp., Armonk, NY). A mixed-design ANOVA, with sex as between-subject factor and time as within-subject factor, was used to investigate changes over time within sexes and possible interactions between time and sex. Where the sphericity assumption was not met, the Greenhouse–Geisser procedure was used to correct the degrees of freedom. A significant interaction between time and sex was followed up with pairwise comparisons with Bonferroni adjustments to compare the mean between different time points for each sex separately. Furthermore, sex differences in absolute changes and calculated percent change from prevalues at each time point were then evaluated with

pairwise comparisons with Bonferroni adjustments for multiple comparisons. If there was no interaction between time and sex, pairwise comparisons between different time points were done for men and women combined. All reported P values from these tests are Bonferroni adjusted. Correlations were investigated with Pearson's r . Missing values were replaced (interpolated) for participants that did not attend one of the posttests because of injury, sickness, or other logistical restraints. Values were calculated by adding the average percentage change for the whole group to the participants' values in the previous test. This was done for one woman at post 1 wk in all biopsy analyses. If a participant lost data at two time points or more for an outcome, the data from this participant were excluded from all time points for this outcome. This was the case for one man at the analyses for HSP and the muscle fiber CSA. Statistical significance was set at an alpha level of 0.05. Values are presented as mean $\pm$ SD.

RESULTS

Body composition. There was a significant main effect of time for body mass ($F(2.1, 32.8) = 40.3, P < 0.001$), muscle mass ($F(3.0, 48.0) = 4.361, P = 0.009$), and fat mass ($F(1.8, 29.4) = 124.5, P < 0.001$), but no significant time–sex interaction in body composition (body mass, $F(2.1, 32.8) = 0.41, P = 0.750$; muscle mass, $F(3.0, 48.0) = 1.3, P = 0.298$; fat mass, $F(1.8, 29.4.0) = 0.42, P = 0.741$).

For the sexes combined, body mass decreased from pre to post 24 h with $5.6\% \pm 1.8\%$ (4.1 ± 1.6 kg, $P < 0.001$) and was still reduced by $2.5\% \pm 2.3\%$ (1.8 ± 1.6 kg, $P < 0.001$) 2 wk after the exercise was finished (Fig. 2A, Table 1).

Mainly all loss in body mass was explained by loss in fat mass, which was reduced by $31\% \pm 11\%$ (3.2 ± 0.9 kg, $P < 0.001$) for the sexes combined 24 h after the exercise compared with pre. Two weeks after the exercise, it was still decreased by $12.5\% \pm 7.7\%$ (1.4 ± 0.9 kg, $P < 0.001$) (Fig. 2C, Table 1).

Only the men decreased muscle mass from pre to 24 h after the exercise with $1.8\% \pm 1.4\%$ (0.7 ± 0.5 kg, $P = 0.026$), and muscle mass was back to prevalues 1 wk after the exercise. There was no significant difference in percentage changes in muscle mass between men and women (Fig. 2E, Table 1).

Leg extension. There were no main effect of time ($F(3.0, 48.0) = 0.489, P = 0.692$) and no significant time–sex interaction ($F(3.0, 48.0) = 0.222, P = 0.881$) in peak isometric leg extension torque as this did not change for neither sex during the study period (Fig. 2B, Table 1).

For peak leg extension torque at $240^\circ \cdot s^{-1}$, there was a significant main effect of time ($F(3.0, 48.0) = 7.423, P < 0.001$) but no significant time–sex interaction ($F(3.0, 48.0) = 0.384, P = 0.765$; Fig. 2D, Table 1).

For the sexes combined, peak leg extension torque at $240^\circ \cdot s^{-1}$ was reduced 24 h after the exercise ($-4.6\% \pm 4.8\%$, $P = 0.012$), 1 wk ($-5.0\% \pm 5.4\%$, $P = 0.011$), and 2 wk ($-4.3\% \pm 4.2\%$, $P = 0.002$) after the exercise compared with pre (Fig. 2D, Table 1).

CMJ. There was a significant main effect of time ($F(3.0, 48.0) = 12.96, P < 0.001$) but no significant time–sex interaction ($F(3.0, 48.0) = 0.287, P = 0.287$) for CMJ.

For the sexes combined, CMJ height was reduced by $6.7\% \pm 6.2\%$ ($P < 0.001$) 24 h after the exercise. It was still reduced by $7.8\% \pm 5.1\%$ ($P < 0.001$) and $4.1\% \pm 4.7\%$ ($P = 0.030$) 1 and 2 wk after the exercise (Fig. 2F, Table 1).

Blood variables. The changes in blood variables are displayed in Table 2. For testosterone, there was a significant effect of time ($F(3.0, 48.0) = 22.5, P < 0.001$) and a significant time–sex interaction ($F(3.0, 48.0) = 21.3, P < 0.001$). The testosterone levels in men decreased by $38\% \pm 26\%$ ($P < 0.001$) 24 h after the exercise. One week after the exercise, it had increased compared with prevalues by $25\% \pm 25\%$ ($P < 0.01$) but was back to prevalues after 2 wk. Testosterone levels in women were as expected low and no changes occurred during the study period.

There was a significant effect of time ($F(3.0, 48.0) = 68.7, P < 0.001$) but no time–sex interaction ($F(3.0, 48.0) = 0.58, P = 0.981$) for IGF1. For both sexes combined, IGF1 decreased during the exercise and was $50\% \pm 9\%$ ($P < 0.001$) lower 24 h after the exercise compared with pre and was still reduced by $11\% \pm 10\%$ ($P = 0.007$) after 1 wk of recovery. After 2 wk recovery, it was back to baseline.

For cortisol, there was a significant effect of time ($F(3.0, 48.0) = 4.1, P = 0.012$) but no time–sex interaction ($F(3.0, 48.0) = 0.35, P = 0.787$). For the whole group, the pairwise comparisons only revealed a tendency to change from pre to 2 wk after the exercise with a $12\% \pm 10\%$ ($P = 0.085$) decrease. However, no changes occurred from pre to 24 h after the exercise. Interestingly, the woman had higher cortisol values than men at every time point ($P < 0.01$).

For CK, there was a significant effect of time ($F(1.7, 24.8) = 20.319, P < 0.001$) but no time–sex interaction ($F(1.7, 24.8) = 1.11, P = 0.355$). For both sexes combined, CK increased by $271\% \pm 180\%$ from pre to 24 h after the exercise ($P = 0.004$). After 1-wk recovery, it was back to baseline.

Muscle fiber CSA and proportions. For muscle fiber CSA, there was no main effect of time for neither muscle fiber type I ($F(2.0, 28.0) = 1.407, P = 0.262$) nor muscle fiber type II ($F(2.0, 28.0) = 1.183, P = 0.321$). There was no time–sex interaction for neither muscle fiber type I ($F(2.0, 28.0) = 0.460, P = 0.636$) nor muscle fiber type II ($F(2.0, 28.0) = 0.170, P = 0.844$) as no changes occurred in neither sex during the exercise or recovery period (Fig. 3A).

For type I muscle fiber proportion, there were no main effect of time ($F(2.0, 30.0) = 0.786, P = 0.465$) and no time–sex interaction ($F(2.0, 30.0) = 0.617, P = 0.564$) as no changes occurred in neither sex during the exercise or recovery period (Fig. 3B).

Calcium-handling proteins. There were no significant main effect of time for SERCA1 ($F(1.4, 22.8) = 2.764, P = 0.098$) or SERCA2 ($F(2.0, 32.0) = 0.065, P = 0.937$) and no time–sex interaction for SERCA1 ($F(1.4, 22.8) = 0.472, P = 0.628$) or SERCA2 ($F(2.0, 32.0) = 0.211, P = 0.811$) (Figs. 4A and 4B).

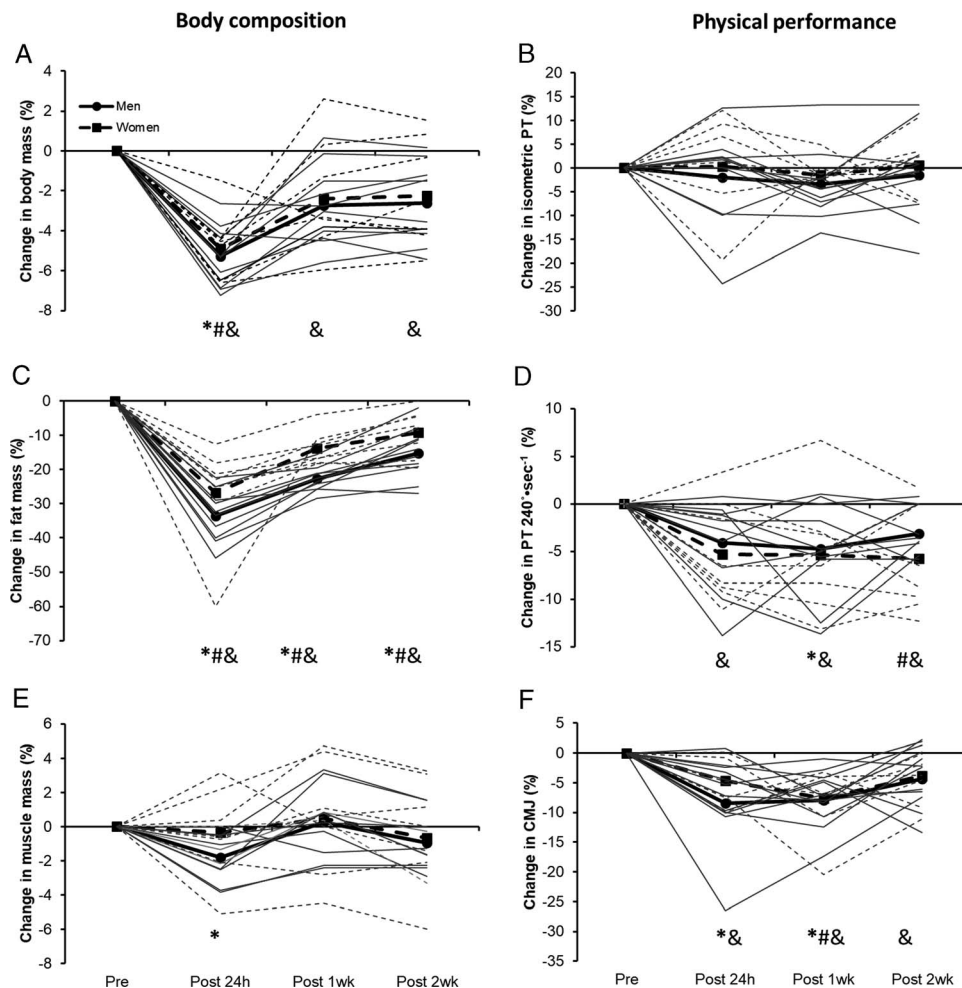


FIGURE 2—Percent changes in body composition (*left column*) and physical performance (*right column*) from before (pre) to 24 h after the exercise (post 24 h) and during the first 2 wk of recovery (post 1 wk and post 2 wk) for men (*solid lines*) and women (*dotted lines*). Thick lines with markers are the group mean, whereas thin lines represent individual changes. Body mass (A); isometric peak torque (PT) (B); fat mass (C); PT at $240^{\circ}\cdot\text{s}^{-1}$ (D); muscle mass (E); countermovement jump (CMJ) height (F). Values are presented as mean $\pm$ SD. * Different than pre ($P < 0.05$) for men. # Different than pre ($P < 0.05$) for women. & Different than pre ($P < 0.05$) for men and women combined.

For RyR1 content in the membrane fraction, there was no main effect of time ($F(2.0, 32.0) = 1.781, P = 0.185$), but there was a significant time–sex interaction ($F(2.0, 32.0) = 5.621, P = 0.008$). The women had an increased content of RyR1 in the membrane fraction 1 wk after the exercise with $45\% \pm 93\%$ ($P = 0.021$), whereas no significant changes occurred in men (Fig. 4C).

No RyR1 band appeared at lower molecular weights, and together with the lack of reduction in the full size RyR1 content, this indicates that no fragmentation of RyR1 occurred.

There were no main effect of time ($F(2.0, 32.0) = 1.576, P = 0.222$) and no time–sex interaction ($F(2.0, 32.0) = 2.157, P = 0.132$) for protein content of calstabin1 in the membrane fraction (Fig. 4D). There was also no main effect of time ($F(2.0, 32.0) = 1.397, P = 0.262$) or a time–sex interaction ($F(2.0, 32.0) = 0.046, P = 0.950$) for the ratio between the membrane protein content of calstabin1 and RyR1. None of these variables changed in neither men nor women during the study period (Fig. 4E).

HSP. For α B-crystallin levels in the cytosolic fraction, there was a significant main effect of time ($F(2.0, 28.0) = 5.399, P = 0.010$) but no time–sex interaction ($F(2.0, 28.0) = 0.72$,

TABLE 1. Body composition and physical performance in response to the cyber endurance exercise and after 2 wk of recovery in both men and women.

Variable		Pre	Post 24 h	Post 1 Wk	Post 2 Wk
BM (kg)	Men	76.7 $\pm$ 6.4	72.6 $\pm$ 6.3*	74.5 $\pm$ 5.6	74.6 $\pm$ 5.8
	Women	65.4 $\pm$ 12.3	62.1 $\pm$ 11.1*	63.6 $\pm$ 10.9	63.8 $\pm$ 11.4
MM (kg)	Men	38.2 $\pm$ 3.6	37.5 $\pm$ 3.4*	38.3 $\pm$ 3.0	37.8 $\pm$ 3.2
	Women	28.8 $\pm$ 3.7	28.6 $\pm$ 3.3	28.8 $\pm$ 3.3	28.5 $\pm$ 3.4
FM (kg)	Men	10.0 $\pm$ 2.6	6.7 $\pm$ 2.1*	7.7 $\pm$ 2.1*	8.5 $\pm$ 2.3*
	Women	14.1 $\pm$ 6.4	10.9 $\pm$ 5.9*	12.2 $\pm$ 5.7*	12.8 $\pm$ 6.0*
CMJ height (cm)	Men	38.0 $\pm$ 8.6	35.1 $\pm$ 9.3*	35.2 $\pm$ 8.9*	36.5 $\pm$ 9.1
	Women	28.8 $\pm$ 4.2	27.4 $\pm$ 4.1	26.4 $\pm$ 3.8*	27.6 $\pm$ 3.6
Isometric PT (N·m)	Men	258 $\pm$ 39	255 $\pm$ 53	250 $\pm$ 45	255 $\pm$ 49
	Women	188 $\pm$ 59	189 $\pm$ 69	186 $\pm$ 64	187 $\pm$ 53
PT $240^{\circ}\cdot\text{s}^{-1}$ (N·m)	Men	121 $\pm$ 21	115 $\pm$ 21	115 $\pm$ 19*	117 $\pm$ 20
	Women	79 $\pm$ 20	74 $\pm$ 19	74 $\pm$ 18	74 $\pm$ 16*

Values are presented as mean $\pm$ SD.

* Significantly different from pre.

BM, body mass; MM, muscle mass; FM, fat mass; CMJ, countermovement jump; PT, peak torque.

TABLE 2. Serum levels of blood biomarkers in response to the selection exercises and after 2 wk of recovery in both men and women.

Variable		Pre	Post 24 h	Post 1 Wk	Post 2 Wk
Testosterone (nmol·L ⁻¹)	Men	18.1 ± 3.5**	10.9 ± 3.6*,**	22.4 ± 5.2*,**	19.7 ± 3.8**
	Women	0.9 ± 0.2	0.8 ± 0.2	1.0 ± 0.3	0.9 ± 0.2
Free testosterone index	Men	6.2 ± 2.2	2.0 ± 0.6*	5.6 ± 1.2	6.6 ± 2.2
	Women	—	—	—	—
SHGB (nmol·L ⁻¹)	Men	31.6 ± 9.5	56.0 ± 13.1*	40.8 ± 9.0*	32.5 ± 10.5
	Women	98.6 ± 45.4	122.8 ± 46.0*	110.2 ± 55.2	98.2 ± 46.6
Cortisol (nmol·L ⁻¹)	Men	574 ± 101**	569 ± 140**	528 ± 98**	481 ± 138**
	Women	749 ± 110	805 ± 162	713 ± 128	682 ± 119
Testosterone/cortisol	Men	3.3 ± 1.1**	2.2 ± 1.4*,**	4.5 ± 1.5*,**	4.6* ± 2.4**
	Women	0.1 ± 0.0	0.1 ± 0.0	0.1 ± 0.0	0.1 ± 0.0
IGF-1 (nmol·L ⁻¹)	Men	29.3 ± 6.9	13.8 ± 2.1*	25.5 ± 6.0*	29.7 ± 7.2
	Women	29.3 ± 7.3	14.6 ± 2.8*	26.2 ± 4.6	30.4 ± 10.2
CK (U·L ⁻¹)	Men	208 ± 109	752 ± 497*	175 ± 145	188 ± 100
	Women	292 ± 321	561 ± 269*	123 ± 54	123 ± 54

Values are presented as mean ± SD.

* Significantly different from pre.

** Significantly different from women at the same time point.

SHGB, sex hormone binding globulin; IGF-1, insulin-like growth factor 1; CK, creatine kinase.

$P = 0.931$). For the sexes combined, α B-crystallin was decreased by $13.5\% \pm 18.9\%$ ($P = 0.024$) 24 h after the exercise and was back to prevalues after 1 wk of recovery (Fig. 5B).

Because of technical difficulties, we were not able to perform α B-crystallin analyses in the cytoskeletal fraction.

There were no main effect of time ($F(2.0, 30.0) = 0.470$, $P = 0.630$) and no time–sex interaction ($F(2.0, 30.0) = 0.835$, $P = 0.444$) for HSP70 protein levels in the cytosolic fraction (Fig. 5D).

For HSP70 in the cytoskeletal fraction, there were a tendency to a main effect of time ($F(1.36, 18.56) = 3.973$, $P = 0.051$) and no time–sex interaction ($F(1.36, 18.56) = 0.833$, $P = 0.405$; Fig. 5E). However, no significant changes in HSP70 during the study was found in the pairwise comparisons.

Fiber type–specific analyses of HSP on muscle cross sections were only possible to investigate in five of the men and four of the women. In these individuals, there was no indication of different HSP responses between fiber types. Therefore, the HSP results are presented independent of fiber type, so we could include all participants also in the HSP analysis.

For α B-crystallin staining intensity in both fiber types combined, there was a significant main effect of time ($F(2.0, 28.0) = 10.032$, $P < 0.001$) but no significant time–sex interaction ($F(2.0, 28.0) = 0.63$, $P = 0.939$).

The group combined tended to increase α B-crystallin staining intensity with $17\% \pm 25\%$ ($P = 0.076$) 24 h after the exercise compared with pre. One week after the exercise, the staining intensity showed no significant different compared with pre values ($-9.4\% \pm 21.7\%$, $P = 0.186$). However, from 24 h to 1 wk after the exercise, there was a significant reduction in staining intensity ($P = 0.002$; Fig. 5A).

The increase in α B-crystallin staining intensity showed a tendency to a positive correlation with the percent change in peak leg extension torque at 240°s^{-1} ($r = 0.454$, $P = 0.077$).

For HSP70, there were no main effect of time ($F(2.0, 30.0) = 0.199$, $P = 0.821$) and no time–sex interaction ($F(2.0, 30.0) = 0.294$, $P = 0.747$) as HSP70 did not change in neither sex during the study period (Fig. 5C).

MuRF1. For MuRF1 protein levels in the cytosolic fraction, there was a tendency to a main effect of time ($F(2, 30) = 2.535$, $P = 0.096$) but no time–sex interaction ($F(1.3, 19.5) = 0.057$, $P = 0.875$).

For both sexes combined, there were no changes from before the exercise to 24 h after exercise, but it tended to decrease with $19\% \pm 35\%$ ($P = 0.072$) 1 wk after the exercise compared with pre values (Fig. 6A).

For MuRF1 protein levels in the cytoskeletal fraction, there was a significant main effect of time ($F(2.0, 30.0) = 9.095$,

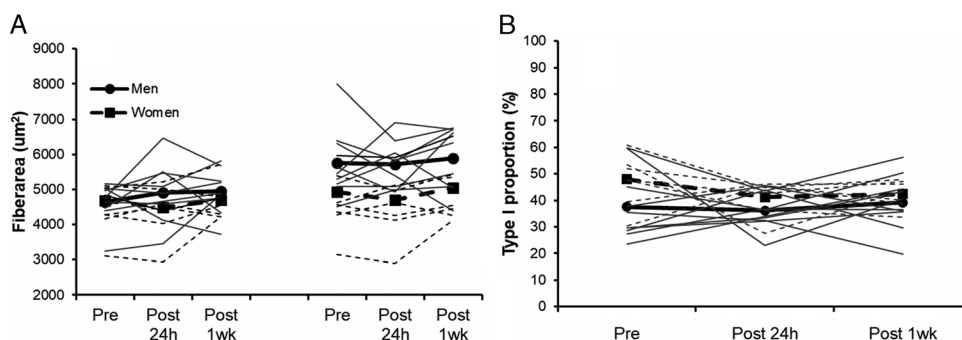


FIGURE 3—Changes in fiber area (A) and fiber type 1 proportion (B) from before (pre) to 24 h after the exercise (post 24 h) and during the first week of recovery (post 1 wk) for men (solid lines) and women (dotted lines). Thick lines with markers are the group mean, whereas thin lines represent individual changes.

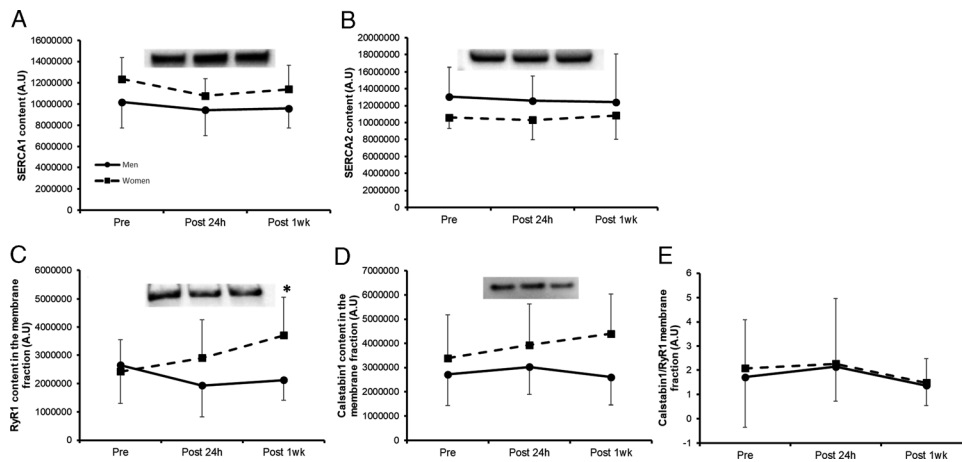


FIGURE 4—Changes in HSP from before (pre) to 24 h after the exercise (post 24 h) and during the first week of recovery (post 1 wk) for men (solid lines) and women (dotted lines). A. α B-Crystallin staining intensity measured by immunohistochemistry. B. α B-Crystallin content in the cytosol fraction. C. HSP70 staining intensity measured by immunohistochemistry. D. HSP70 content in the cytosol fraction. E. HSP70 content in the cytoskeleton fraction. # Tendency to different than pre for the men and women combined ($P < 0.1$).

$P < 0.001$) but no time–sex interaction ($F(2.0, 30.0) = 1.220$, $P = 0.309$)

For both sexes combined, there were no changes in MuRF1 protein levels in the cytoskeletal fraction from before the exercise to 24 h after the exercise. However, 1 wk after the exercise, the levels had increased with $85\% \pm 105\%$ compared with pre ($P = 0.003$; Fig. 6B).

DISCUSSION

In line with our hypotheses, the 10-d demanding military field exercise had larger negative effects on the ability to produce force at high contraction velocities and explosive strength compared with isometric strength, as peak leg extension torque at 240°s^{-1} and CMJ height were significantly impaired, whereas no impairment was observed in isometric MVC leg extension torque. Furthermore, the ability to produce force at high contraction velocities was not recovered in the 2-wk recovery period. Contrary to our hypotheses, these findings could not be explained by any changes in the investigated

proteins involved in the Ca^{2+} -handling systems, changes in muscle fiber composition, or fiber type-specific changes in fiber area and indicators of muscular stress. However, there was increased content of MuRF1 in the cytoskeleton fraction 1 wk after the exercise. There were no observed sex differences in changes in physical performance, nor in changes in the muscle cells. The military field exercise led to a similar reduction in body mass in both men and women, and this reduction was mainly caused by reduction in fat mass. Nevertheless, a slight reduction in muscle mass was observed in the men. The changes in blood biomarkers followed the expected changes related to energy deficiency and demanding exercise and did not differ between sexes. All alternations in blood biomarkers recovered within 1 wk.

The reductions in explosive strength in the legs are similar to some previous studies (1,6,9,13) and smaller than others (4,10) of comparable length (7–14 d). The lack of change in maximal strength measured as isometric torque in leg extension is also in agreement with some studies (27) but contradict other studies (4,13). The main reason for the conflicting

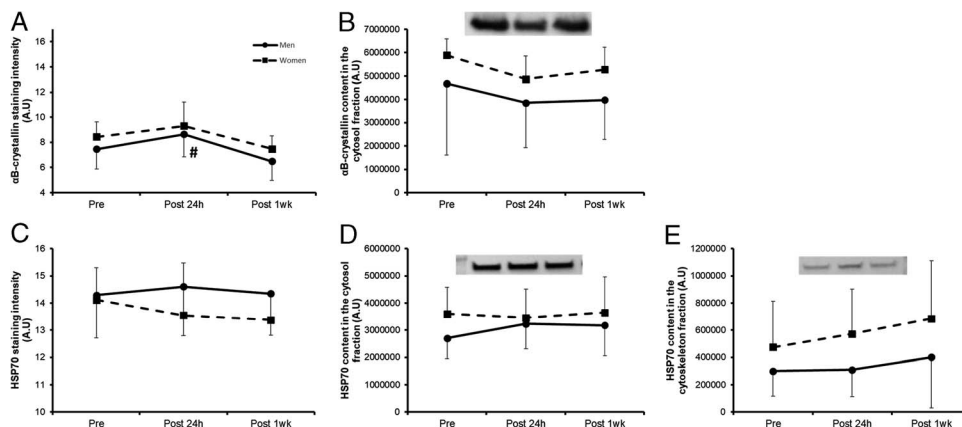


FIGURE 5—Changes in protein related to calcium handling from before (pre) to 24 h after the exercise (post 24 h) and during the first week of recovery (post 1 wk) for men (solid lines) and women (dotted lines). A. SERCA1 content. B. SERCA2 content. C. RyR1 content in the membrane fraction. D. Calstabin1 content in the membrane fraction. E. Calstabin1/RyR1 ratio in the membrane fraction. * Different than pre ($P < 0.05$) for men or women.

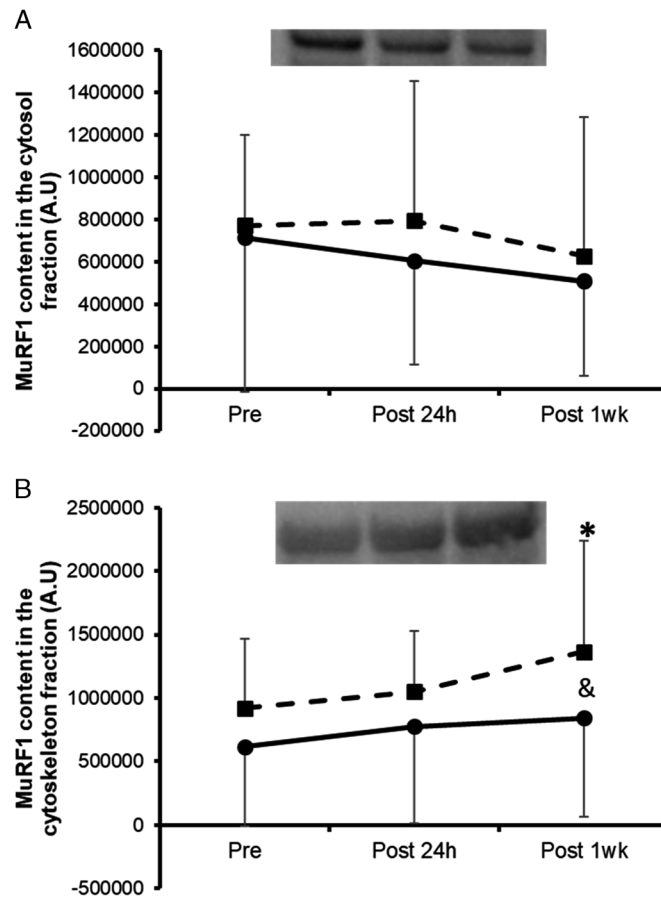


FIGURE 6—Changes in protein levels of MuRF1 from before (pre) to 24 h after the exercise (post 24 h) and during the first week of recovery (post 1 wk) for men (solid lines) and women (dotted lines). A. MuRF1 content in the cytosol fraction. B. MuRF1 content in the cytoskeleton fraction.

findings regarding physical performance in different studies is probably related to differences in the field exercises. Studies with large reductions in performance typically include higher activity levels (6000 to 7000 kcal·d⁻¹), more demanding physical activity, lower energy intakes (~550 kcal·d⁻¹), and hence a more negative energy balance (4,10), whereas studies with similar reduction have approximately the same energy consumption, activity levels, and energy intake as the current study (1,9,13). Despite the modest stress caused by the exercise and energy deficiency, the recovery from the impairments in explosive strength and ability to develop force at high contraction velocities was prolonged and not fully recovered after 2 wk of recovery. Prolonged recovery has also been reported after the more demanding field exercises described above (4,10). This indicates that even military field exercises with moderate acute reductions in performance can still lead to long-lasting reductions in explosive strength and muscular power. Such long-lasting impairments in muscle power-generating capacity should be considered when planning military training and operations. The similar changes between men and women in physical performance during the exercise were expected based on the limited amount of previous studies with (10). However, in contrast to our previous study where women recovered faster in CMJ height and maximal power than men (10), there were no sex differences in the recovery process of

physical performance in the present study. The reason for the different findings is unclear but might be related to the much larger declines in performance in our previous study caused by a more extreme field exercise with a much higher energy expenditure (~90 vs ~75 kcal·kg⁻¹·d⁻¹) and lower energy intake (~550 vs ~900 kcal·d⁻¹).

An interesting finding from the current study is that the demanding field exercise had no negative effect on maximal isometric torque but led to a long-lasting decline in the ability to produce force at high contraction velocities and explosive strength. We measured the power-related abilities with the CMJ but also with the peak leg extension torque at 240°·s⁻¹. This was because we wanted to measure power using a non-complex exercise involving the same muscles and movement as the isometric test to rule out that reductions in CMJ performance were mainly related to other factors, such as pain or technique. Our finding that explosive strength and power-related measures are more affected than other aspects of physical performance after military field exercises is also supported by previous studies (4,10,13,27) and a recent meta-regression (28). However, the physiological mechanisms behind the prolonged impairments in specific aspects of muscle function after military field exercises are unclear.

It has been speculated that the observed changes in body composition and in anabolic and catabolic hormones could

explain changes in physical performance (2,3,7,29). This is not supported in our results as there were generally small to no correlations between the changes in muscle mass, endocrine changes, and changes in performance during the exercise, and all analyzed hormones had recovered after 1 wk of recovery when physical performance was still depressed. Furthermore, only men lost muscle mass during the exercise, whereas the changes in performance were similar in men and women. The lack of relationship between changes in body composition and performance has been demonstrated previously by our group, where large sex differences in muscle mass were not accompanied by any differences in physical performance (10).

The reductions in body mass and fat mass after the field exercise in the current study were similar (9) or lower (10) than the reductions reported in field exercises of comparable duration. The differences between studies can be explained by variations in energy expenditure and energy intake. Studies with larger changes in body composition than our study have in general included more demanding field exercises with larger energy deficits (4,7,10). In the current study, almost all change in body mass could be explained by changes in fat mass in both sexes. However, some of the changes in body mass might also be because of dehydration and low muscle glycogen stores after the excises. To minimize the effect of dehydration, we did the postmeasurement 24 h after the exercise under the same standardized conditions as on the other time points. The lack of change in muscle mass in the female soldiers is in accordance with a previous study from our research group (10) and further supports the notion that women have a high ability to preserve muscle mass during demanding military field exercises. The almost no change in muscle mass in men and the lack of sex differences in changes in muscle mass are not in accordance with most studies investigating the effects of demanding field exercises in men (2,4,7) and previous studies comparing men and women (5,10). A possible explanation for the different findings might be related to the content of the field exercises, for example, the size of the negative energy balance. In our previous study, activity level and energy expenditure were larger (~ 90 vs ~ 75 kcal $\cdot$ kg $^{-1}\cdot$ d $^{-1}$) and energy intake lower (~ 550 vs ~ 900 kcal $\cdot$ d $^{-1}$), leading to a much larger energy deficit than in the current study. Another possible explanation might be differences in initial fat mass of the participants as the male participants in our previous study had very low initial mass, whereas we observed normal initial fat mass in the male participants in the current study.

The reduction in the anabolic hormones, testosterone and IGF-1, confirms that the soldiers were in a catabolic state, which is in good agreement with the changes in body mass and fat mass. The changes in anabolic hormones are similar to reports in studies of comparable length and with similar negative energy balance (13,30,31). In more challenging (e.g., higher activity levels) or exercises with longer duration and larger total energy deficit, even larger reduction has been reported (7,10). The changes in anabolic hormones are believed to mainly be related to the negative energy balance and to a lesser degree to the lack of sleep (29).

Because explosive strength and power-related measures appear to be more affected than other aspects of performance, the decline in performance should partially be related to mechanisms that affect the ability to produce force at high contraction velocities more than isometric force. One possible mechanism might be changes in muscle fiber composition or a greater negative effect on the type II than type I fibers. No previous studies have investigated fiber type-specific changes after a military field exercise. However, studies investigating the effects of short-term hypocaloric diets report a larger muscle fiber atrophy in type II fibers than type I fibers in both rodents (32) and humans (33). One study in rats reported an overall fiber type shift from fast fibers toward slow fibers can last for up to 2 wk (34). Furthermore, such hypocaloric diets have been reported to reduce the maximal rate of force development, to slow the maximal relaxation rate, and to increase losses of force at high compared with low stimulation frequencies. (34,35). However, in the current study, there were no changes in fiber area in neither type I nor type II fibers and no changes in fiber type composition. Therefore, a larger atrophy in type II than type I fibers and a fiber type shift toward a slower fiber type cannot explain the reduced explosive strength in the current study.

Hamarsland and colleagues (4) suggested that damage to the contractile apparatus in muscle fibers is an important mechanism for impairment of physical performance. The increase in CK levels after the exercise in our study indicates that some muscle damage was present. However, the increase in CK values after the exercise was moderate compared with previous studies (4,10), and none of the soldiers had indications of rhabdomyolysis. There was also no correlation between changes in CK values and reduced performance. Interestingly, α B-crystallin levels in the cytosolic fraction were reduced 24 h after the exercise together with a tendency to increase staining intensity in the muscle sections. The biological role of α B-crystallin and other small HSP is to act as protein chaperones and interact with denatured and unfolded proteins (18). Small HSP has cytoprotective properties as they translocate and accumulate in the cytoskeletal/myofibrillar fraction after muscle damaging activity (16). Increased staining on muscle section has been shown to reflect an accumulation in myofibrillar structures (16,18). Consequently, the decreased levels in the cytosolic fraction together with the tendency to increased staining indicate that some translocation also occurred after the field exercise in the current study. Both eccentric muscle contractions and other forms of strenuous muscle exercise have been reported to result to translocation of small HSP from the cytosolic fraction to the cytoskeletal/myofibrillar fraction and increased synthesis of these protein the following days (16,18,19). Increased staining for α B-crystallin and decreased levels in the cytosol in the current study therefore further support that the muscle cells experienced some stress and damage during the field exercise. However, the increased staining intensity for α B-crystallin was in most cases evenly distributed across the muscle fibers and the punctuated and granular staining, which is considered a marker for myofibrillar disruption

that occurs after eccentric and heavy load strength training (18,36) was not observed. Therefore, the increased α B-crystallin staining in the current study probably does not indicate large disruptions of the myofibrillar structure but some type of other cellular stress that could affect performance. The exact type of stress indicated and what structures are affected is unclear, but the staining pattern is similar to what has been reported after blood flow restricted exercise (19). Interestingly, acute aerobic nondamaging exercise (running) has been reported not to affect total expression of α B-crystallin 24 h to 7 d after exercise, although translocation was not measured (37). It should be noted that we took our biopsies after a 10-d long field exercise, and we do not have a clear overview over when in this period the possible muscle damaging work occurred or if the stress was evenly distributed during these 10 d. Therefore, we do not know how long after the damaging activity the biopsy was taken although the energy consumption was very similar from day to day, indicating that the stress was quite evenly distributed. Because the staining of α B-crystallin appears to decline quite rapidly from 48 h after muscle damaging eccentric exercise (16), this makes it challenging to compare our results to those of others.

A tendency for a moderate positive correlation was found between the percent change in staining intensity of α B-crystallin and peak torque during isokinetic contraction at $240^{\circ}\cdot\text{s}^{-1}$. This is not in accordance with a study reporting a large negative correlation between percentage fibers positive with small HSP (HSP27) and change in force generation capacity after unaccustomed muscle damaging eccentric exercise (18). However, after this kind of exercise, larger acute accumulation of small HSP probably represents muscles with larger muscle damage and therefore correlates negatively with change in force generating capacity. The situation is quite different in the current study where the amount of damaging exercise probably was much less and hence tolerable for the participants. Increased levels of different HSP have actually been speculated to lead to increased protection of myofibrillar structures during repeated exposure to maximal eccentric exercise (16). Thus, it might be speculated that the cytoprotective role of α B-crystallin protected the subjects with higher accumulation in the muscle fiber better against muscle damage from the exercise, which consequently results in less reduction in force generating capacity.

We did not see any significant changes neither in the HSP70 measured by IHC nor in the cytosolic and cytoskeletal fraction measured by Western blotting. This is surprising because increased staining intensity of HSP70 in muscle cross section and increased levels in the cytosolic and the cytoskeleton fraction have been reported after various modalities, including aerobic nondamaging exercise, BFR exercise, and muscle damaging eccentric exercise (16,18,19,37). However, studies in well-trained rowers have shown that HSP70 expression is higher after high-intensity exercise compared with lower intensity. Additionally, low-intensity endurance training does not always lead to increased HSP70 expression (38). This might explain our results because the field exercise mostly

had low-intensity prolonged activity with some amounts of high intensity. Another explanation for the lack of changes in HSP70 levels can be related to the timing of the biopsies. We collected the muscle samples 24 h and 1 wk after the exercise. Because the exercise lasted for 10 d, we cannot rule out the possibility that high activity during the first days increased HSP70 levels that returned to preexercise values before the first postbiopsy. Depending on type of activity, the typical HSP70 response after acute sessions usually consists of none to small changes during the first hours, before a more substantial increase after approximately 24 h, and reach a peak after approximately 48 h (18,19), with quite large individual differences. However, to the best of our knowledge, no study has investigated the time course for the HSP response after several days with high daily energy expenditure, such as a military field exercise.

Type II fibers have been reported to be more prone to muscle damage after eccentric work than type I fibers (16), and we aimed to investigate if the same could be seen after a military field exercise and whether this could be a possible explanation for the larger reduction in explosive strength compared with maximal strength. However, this was measured in only five men and four women so caution is warranted. There is a tendency to increase in α B-crystallin staining, and no changes in HSP70 staining were similar in type I and type II fibers, suggesting that both fiber types experienced similar stress. This is in contrast to a predominantly increased α B-crystallin staining in type I fibers after BFR exercise (19) and a stronger staining of HSP27 in type II fibers after traditional heavy strength training (36).

We found a tendency to increased protein levels of MuRF1 24 h after the exercise and a large increase 1 wk after the exercise in the cytoskeletal fraction. This was accompanied with a tendency to a small reduction in the cytosolic fraction 1 wk after the exercise. MuRF1 is an E3 ligase specifically expressed in muscle (39), and among its known targets are several myofibrillar proteins, including troponin 1, myosin heavy chains, actin, and myosin light chains (39). The specific increase in the cytoskeletal fraction indicates MuRF1 binding to and ubiquitinating myofibrillar proteins, targeting them for destruction by the 26S proteasome in the week after the exercise. However, caution is warranted because we did not measure total ubiquitination or activity in any of the subunits of the 26S proteasome. Furthermore, as discussed, we did not find any reduction in muscle fiber CSA.

Several studies have reported acute changes in gene expression and protein levels of MuRF1 after both strength and endurance exercise (40–42), but only a few studies have investigated this after military field exercises or other similar challenges. A previous study investigating the effects of a similar military field exercise on the activation of the ubiquitin–proteasome system reported a very small reduction of MuRF1 protein levels in muscle homogenate but no changes in measures of proteasome activity 1–2 h after the exercise (43). We found the biggest changes 1 wk after the exercise, so with the different timing of the biopsies, it is difficult to compare with our

study. However, as the homogenization procedure used by Moberg et al. (43) does not include cytoskeletal proteins in the supernatant, the results are somewhat similar to our finding of a reduced amount in the cytosolic fraction. Ultra-endurance running has been reported to lead to an acute increase in the mRNA expression and protein levels of MuRF1 together with increased proteasome activity in muscle homogenates (44,45). Furthermore, 21 d of energy deficit caused by an reduction in energy intake and increased energy expenditure also increased mRNA levels of MuRF1 (46). These studies indicate that a negative energy balanced caused by a high activity levels combined with reduced energy intake can led to the breakdown of muscle protein through increased levels of MuRF1 and the activation of the ubiquitin–proteasome system. Timing wise, the increased MuRF1 in the cytoskeletal fraction corresponds to the reduced physical performance, and MuRF1 has been reported to be fiber type–dependent and is preferentially induced in type II fiber atrophy (47). This correspond well with the larger reduction of explosive strength compared with isometric strength in the current study. Therefore, it can be speculated that the breakdown of myofibrillar proteins plays a role in the prolonged reduced explosive strength reported in the current study, and this should be further investigated in future studies.

Another possible mechanism behind the reduced explosive strength might be related to impaired Ca^{2+} handling and Ca^{2+} dynamics in the muscle cells. Reduced Ca^{2+} release due to changes in RyR1 content or function or alterations in SR Ca^{2+} content due to changes in SERCA content and/or function could cause reductions in explosive strength. However, there were only small to no changes in the content of RyR1, SERCA1, SERCA2, and calstabin1 and no sign of any dissociation of calstabin1 from RyR1 or fragmentation of RyR1. When the channel-stabilizing calstabin1 dissociates from RyR1, it can cause Ca^{2+} to leak and impair muscle function and fatigue resistance (20). The lack of dissociation in the current study was unexpected because this dissociation has been reported after various exercise modalities, including prolonged periods of high activity, marathon running, and high-intensity endurance training in both humans and rodents (20,21,48). However, to the best of our knowledge, this is the first study investigating this dissociation of calstabin1 from RyR1 after a military field exercise consisting of high daily energy expenditure. It might be that the activity patterns and periods of lighter engineering task were too modest to induce a dissociation. Although this mechanism cannot be excluded for exercises with larger reductions in performance (10), it is therefore unlikely to be the primary cause for the reduced performance observed in this study. The physical demands to enter the Norwegian Defense Cyber Academy are quite high, and physical training is an important part of the soldier's education. Therefore, the soldiers participating in this study were quite well trained, and it might also be speculated that their training status prevented any dissociation as it has been shown that this dissociation and other modifications of RyR1 is less pronounced as training status increases (48). However, a dissociation has been reported in trained cyclists after a 3-h cycling protocol

at 70% of $\dot{V}\text{O}_{2\text{max}}$ and after a marathon in athletes at a recreational level (20). Although a detailed registration of the soldiers' training habits was not collected, it is unlikely that their training status was too high to induce dissociation. Most previous studies investigating the dissociation of calstabin1 from RyR1 have used the immunoprecipitation of the RyR1 complex (20,21,48). Unfortunately, that method was not possible to use in this study. Therefore, this was analyzed in the membrane fraction where RyR1 mainly can be found. With the assumption that the dissociated calstabin1 will end up in the cytosolic fraction, a dissociation of calstabin1 should lead to a reduced calstabin1 and/or reduced calstabin1/RyR1 ratio in the membrane fraction. However, this method has not been verified; hence, the used method may be not suitable for detecting this dissociation.

A fragmentation of the RyR1 complex has been reported to cause the channels to become leaky at rest, leading to increased resting cytosolic Ca^{2+} , impaired Ca^{2+} release, and depressed muscle force (21). Therefore, a similar fragmentation after demanding military field exercises could be related to the reduced explosive strength and the ability to develop high power. However, there was no sign of RyR1 fragmentation. RyR1 fragmentation has been previously reported after high-intensity interval training consisting of 30-s all-out efforts (21,49,50), but not after more prolonged exercise such as a marathon (21). Thus, it appears that RyR1 fragmentation is more likely to occur after maximal-intensity, short-term exercise, as HIIT, rather than prolonged low-intensity activity such as typical for the field exercise conducted in this study

Limitations

The fact that both sexes performed the field exercise together in mixed teams ensured that the physical demands of the exercise were as similar as possible between men and women. However, we do not know how each team distributed their load carriage and other tasks, so we cannot rule out the possibility that in some or all teams, men were chosen to carry the heaviest equipment and perform more physical strenuous tasks. To get a rough estimate of differences between the sexes in the energy expenditure during the field exercise, we estimated it using wrist-worn accelerometers. The energy expenditure was higher in men in absolute values, but no sex differences were discovered when normalized to body mass. These data suggest that men and women experienced similar relative levels of physical exertion during the exercise. However, the accelerometers do not account for the load carried by the soldiers so we still cannot rule out differences in relative loads.

The study has a quite low sample size that might affect our outcomes, especially when comparing sexes. This makes it difficult to draw conclusions regarding sex differences, especially for the muscle biopsy outcomes where the variation is generally larger. However, in most of our results, data from men and women are combined, increasing the sample size and power.

As discussed earlier, we do not have a clear overview over when in the 10-d long field exercise the heaviest work was done and therefore does not have a complete overview of

when compared with the hardest part of the exercise the post-tests are done.

CONCLUSIONS

This is the first study to investigate the effects of a demanding military field exercise on muscle cellular outcomes and its role in explaining the prolonged impairments in physical performance after such exercises, and possible sex differences in this regard. The study confirms that explosive strength and the ability to generate force at high contraction velocities are the parameters that are most affected by demanding military field exercises and takes the longest time to recover in both sexes. Changes in these abilities were not recovered 2 wk after the moderately demanding field exercise.

These reductions in performance were not related to the investigated changes in Ca^{2+} handling, changes in muscle fiber

composition or fiber type-specific changes in fiber area, or muscular stress measured by translocation and increase in HSP. However, the increased levels of MuRF1 in the cytoskeleton fraction 1 wk after the exercise might be related to the reduced explosive strength. Furthermore, unlike our previous study, we did not observe any sex differences in body composition or physical performance during the exercise or recovery period, and the lack of sex differences may be explained by the less strenuous activity resulting in only modest alterations in both sexes.

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